

Create an R script for data generation

Sidney Reynaud

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```
# Step 1: Load required packages
library(dplyr)

## Warning: package 'dplyr' was built under R version 4.3.3

##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union

# Step 2: Set seed and define sample size
set.seed(123)
n <- 500

# Step 3: Generate synthetic dataset
synthetic_data <- data.frame(
  PageViews = rpois(n, lambda = 5),
  TimeSpent = rexp(n, rate = 1/7),
  Version    = sample(c("A", "B"), size = n, replace = TRUE),
  Conversion = NA_integer_
)

# Step 4: Assign conversion probabilities by version
synthetic_data$Conversion[synthetic_data$Version == "A"] <-
  rbinom(sum(synthetic_data$Version == "A"), size = 1, prob = 0.10)

synthetic_data$Conversion[synthetic_data$Version == "B"] <-
  rbinom(sum(synthetic_data$Version == "B"), size = 1, prob = 0.15)

# Step 5: Save synthetic dataset to CSV
write.csv(
  synthetic_data,
  "practical-application/synthetic_data.csv",
  row.names = FALSE
)
```